Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Medical · Class 11

Subject: Chemistry

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Chemical Bonding: Ionic and Covalent Bonds

DM

Unit 1: Chemical Bonding: Ionic and Covalent Bonds

Learning Objectives

• Students will explain why atoms form chemical bonds in terms of achieving a stable electron arrangement.

• Students will distinguish ionic bonds from covalent bonds using electron transfer and electron sharing.

• Students will predict the type of bond in a compound from the elements involved and relate bond type to physical properties.

Key Concepts

• Atoms bond to achieve a stable octet of electrons in their outer shell, the same arrangement as a noble gas.

• An ionic bond forms when a metal atom transfers one or more electrons to a non-metal atom, creating oppositely charged ions that attract each other; sodium chloride is the classic example.

• A covalent bond forms when two non-metal atoms share one or more pairs of electrons, as in a hydrogen molecule or a water molecule.

• A single covalent bond shares one pair of electrons, a double bond shares two pairs and a triple bond shares three pairs.

• Ionic compounds are usually solids with high melting points and conduct electricity when molten or dissolved, while covalent compounds are often gases, liquids or low-melting solids.

Teaching Methodology (Step by Step)

1. The teacher asks why noble gases rarely react and links their full outer shells to the idea of stability that drives bonding.

2. The teacher draws the electron transfer from sodium to chlorine and shows how Na+ and Cl- ions attract to form sodium chloride.

3. The teacher contrasts this with two hydrogen atoms sharing one pair of electrons to form H2, drawing the shared pair between them.

4. The teacher builds up to double and triple bonds with oxygen (O2) and nitrogen (N2) dot-and-cross diagrams.

5. The teacher compares the properties of sodium chloride and water and asks students to link each property to its bond type.

6. The teacher gives five compounds and asks students to predict ionic or covalent bonding and justify each choice.

Classroom Activities

• Bond sort: students classify a list of ten compounds as ionic or covalent and give a reason for each classification.

• Dot-and-cross draw: in pairs, students draw dot-and-cross diagrams for NaCl, H2O and CO2.

• Property match: students match properties such as high melting point and poor conductivity to the correct bond type.

Assessment Questions

1. Explain, with a diagram, how an ionic bond forms between magnesium and oxygen.

2. Why is water a covalent compound while sodium chloride is ionic?

3. Predict whether calcium fluoride is ionic or covalent and state two properties you would expect it to have.

Homework

• Draw dot-and-cross diagrams for hydrogen chloride (HCl) and magnesium chloride ($MgCl_2$) and label the bond type in each.

• List four differences between ionic and covalent compounds with one example of each.

Time Allocation

40 minutes

DM